Research Article

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Effects of speech-enabled LLM-based chatbots on L2 learning outcomes and affective domains: a multilevel meta-analysis

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Abstract: Recent developments in natural language processing and automatic speech recognition have significantly advanced chatbot communication using the voice mode, establishing these tools as viable resources for L2 speaking practice. More recently, the L2 chatbot domain has leveraged large language models (LLMs) to enhance contextual understanding and facilitate meaningful dialogues, prompting research into the effects of speech- and LLM-based chatbots on L2 learning outcomes and affective domains. This study presents a synthesis of these findings using a multilevel meta-analysis. Based on a dataset of 19 studies (32 samples, N = 2,643), the results revealed that the overall effect size for L2 learning outcomes was 0.68 ($p < 0.001$). The overall effect size for affective domains was 0.64 ($p < 0.001$). Moreover, moderator analyses indicated that a range of learner and intervention characteristics, including learner proficiency, educational level, interaction format, and context of chatbot use, significantly influenced the overall effectiveness.

Keywords: speech-enabled chatbot; generative AI; L2 oral communication; multilevel meta-analysis; affective variable

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1 Introduction

This study reports a multilevel meta-analysis of the effects of speech-enabled, large language model (LLM)-based chatbots (henceforth, speech-LLM chatbots) on second language (L2) learning outcomes and affective domains. Chatbots, the technology at the center of this synthesis, are "computer programs that use AI to simulate human conversation" (AbuSahyon et al. 2023: 1212). In language teaching and learning, they have long been considered useful practice tools for target language (TL) learners (Atwell 1999; Fryer and Carpenter 2006; Fryer et al. 2020; Jeon et al. 2024; Lee et al. 2020). Although chatbots offer several distinct affordances in TL classes (Huang et al. 2022), their function as a conversational partner has received the most attention (Jeon et al. 2024), largely because language classrooms typically lack sufficient opportunities for communication practice in the TL (Shadiev and Liu 2023). In what follows, we outline the pedagogical merits attributed to chatbots and the technological developments that have produced speech-LLM chatbots, and then review prior syntheses to identify the gap the present study addresses.

In the educational literature, chatbots have been suggested to have several pedagogical merits. Based on a review of 46 articles on chatbots employed for pedagogical purposes across various domains, Zhang et al. (2024b: 4541) proposed the RAISE model of chatbot-assisted learning, which identifies such merits as "Repetitiveness, Authenticity, Interactivity, Student-centredness, and Enjoyment". Chatbots enable the implementation of repetitive tasks and practice while providing a more authentic learning environment through opportunities for realistic conversation. The technology is also highly oriented toward interactivity (i.e., students can engage in extensive and rich interactions) and student-centredness, allowing students to regulate their learning pace and content. Finally, chatbots provide an enjoyable learning experience through the use of multimedia, emojis, and embodiments, enhancing motivation and encouraging continued use of this tool. Although Zhang et al. (2024b) include domains other than language learning, the extant research and systematic reviews within the language learning field have proposed similar pedagogical affordances (Fryer et al. 2020; Huang et al. 2022; Jeon et al. 2024; Lee et al. 2020).

While chatbots were originally developed as text-mode tools (e.g., ELIZA in Weizenbaum 1966), and such text-based chatbots were examined extensively in earlier language learning studies (Fryer and Carpenter 2006; Kim 2017), recent advancements in automatic speech recognition (ASR) (Oh and Song 2021; Shadiev and Liu 2023) and natural language processing (NLP) (Liang et al. 2023) have led to the development of speech-based chatbots, prompting numerous relevant language learning studies (Lee and Jeon 2024; Timpe-Laughlin and Dombi 2020; Yang et al. 2022). More recently, the integration of LLMs has made conversations with chatbots even more authentic, and

research on such chatbots has become one of the major strands of recent technology-mediated TL research (Boudouaia et al. 2024; Darmawansah et al. 2025; Escalante et al. 2023; Ghafouri 2024; Liu et al. 2026; Tsai et al. 2024; Zou and Huang 2024). Generative AI (GenAI) chatbots powered by LLMs are significantly more advanced than earlier iterations, as they "can connect contexts, understand complex language structures and contexts, and thus have fluent and natural conversations with learners by logical and complete sentences" more effectively (Li et al. 2025: 9). This convergence of speech technology and LLMs has given rise to the speech-LLM chatbots examined in this study, raising the question of how effective these tools actually are for TL learning.

Given their status as a primary pedagogical resource in modern TL teaching and learning, systematic efforts have been dedicated to evaluating chatbot effectiveness and pedagogical affordances. Bibauw et al. (2022) conducted a meta-analysis of 17 studies on the effects of dialogue-based systems (including voice- and text-mode chatbots) on TL learning; their multilevel random-effects model revealed a medium effect size ($d = 0.58$) for TL proficiency development, and moderator analyses indicated that certain features, such as form-focused systems, system-guided interactions, and various forms of gamification, may lead to higher TL gains. Huang et al. (2022) analyzed 25 empirical studies on voice- and text-mode chatbots, identifying various technological (e.g., temporal and spatial flexibility, ease of use), pedagogical (e.g., simulating authentic learning environments, recommending materials), and social (e.g., interpersonal and cohesive communication) affordances for TL learning. Turning to speech-enabled chatbots specifically, Jeon et al. (2024) identified 32 empirical studies and revealed that the most common role of such chatbots was that of a conversational partner, with learners' perceptions and their development of TL speaking or listening being the most extensively examined domains. Notably, however, both this review and Shadiev and Liu (2023) synthesized speech-recognition and speech-enabled chatbot research qualitatively, mapping roles, affordances, and research foci, whereas the broader quantitative syntheses of chatbot-assisted language learning (e.g., Bibauw et al. 2022; Li et al. 2025; Lyu et al. 2025; Wang et al. 2025; Wu and Li 2026) have pooled voice- and text-mode tools together, typically without isolating LLM-based spoken interaction. To our knowledge, no quantitative synthesis has been restricted to LLM-based chatbots, included only studies involving genuine two-way oral communication, and modeled learning and affective outcomes separately within a multilevel framework – precisely the gap the present study addresses.

1.1 Interpretive framework

This work draws on an interactionist view of second language acquisition, in which comprehensible input, meaningful output, and interactional feedback drive L2

development (Long 1996; Swain 2005), complemented by the Control–Value Theory of achievement emotions (Pekrun 2006, 2024; see Shao et al. 2019 for its application to L2 learning), which frames these emotions as products of learners' appraisals of control over, and value attached to, a learning activity. From the interactionist perspective, speech-LLM chatbots function as tireless conversation partners supplying spoken input, eliciting oral output, and providing immediate, face-saving feedback; from the Control–Value perspective, their low-stakes, responsive nature bears directly on learners' perceived control and, through it, on anxiety. This dual framework is also consistent with the decision, which follows earlier chatbot meta-analyses (e.g., Lyu et al. 2025), to model learning and affective outcomes as distinct domains. We adopt it as an interpretive lens rather than a source of a priori predictions: the moderators were not derived from it but from prior chatbot meta-analyses (Section 2.2.3), and the framework serves to organize and interpret them. Within this framework, learner-related variables (e.g., L2 proficiency, educational level) index the appraisals and processing capacities shaping how learners benefit from interaction, whereas intervention-related variables (e.g., interaction format, modality, task type, context of use) index the quantity, quality, and framing of the interaction itself.

Control–Value Theory also specifies the architecture linking the affective constructs examined here. In Pekrun's (2006, 2024) model, appraisals of controllability (e.g., self-efficacy) and value jointly generate achievement emotions such as enjoyment and anxiety, which in turn shape motivation, engagement, and willingness to communicate (Dewaele and MacIntyre 2014; MacIntyre et al. 1998; Pekrun and Linnenbrink-Garcia 2012). The constructs synthesized in this meta-analysis (i.e., anxiety, enjoyment, motivation, engagement, self-efficacy/self-confidence, and willingness to communicate) thus occupy distinct positions within a single appraisal–emotion–motivation system. They are related but not interchangeable; the implications of this distinction for pooling them are discussed in Section 2.2.2.

The framework also suggests mechanisms through which each moderator may operate. Among learner-related variables, L2 proficiency may matter because LLM-based chatbots can adapt their output and repair breakdowns: lower-proficiency learners, who often struggled with earlier, rigid speech-recognition systems (Wang et al. 2025), may realize disproportionate gains. Developmental maturity (educational level) may in turn condition the control and value appraisals learners bring to autonomous technology use (Eccles and Wigfield 2020; Pekrun 2024). Among intervention-related variables, interaction format may alter the conversational load and the social character of appraisals (Lee 2025; Tai and Chen 2024); modality may affect the balance of input channels (Lyu et al. 2025); intervention length and context of use bear on the quantity of interaction and degree of learner control, which Control–Value Theory treats as an environmental antecedent of control appraisals (Loderer et al. 2020; Pekrun 2006); and task type may

moderate learning gains (cf. Bibauw et al. 2022), partly by shaping instructional framing and hence perceived value (Guo et al. 2026). These mechanisms structure the interpretation of the moderator results (Section 4).

1.2 The present study

Against this backdrop, we conducted a meta-analysis of empirical studies on the effectiveness of speech-LLM chatbots for TL learning, considering both learning outcomes and the affective domains. A meta-analysis is especially warranted here because the primary evidence is not only sparse but also mixed. While many studies report sizeable gains in oral proficiency and positive affective shifts (e.g., Kooti et al. 2025; Wu et al. 2026), others find non-significant or negligible differences between chatbot and comparison conditions in certain domains (e.g., Sonsaat-Hegelheimer and Kurt 2024), and effects appear to vary with learner and intervention characteristics. When individual experimental findings are inconsistent in this way, a quantitative synthesis is better suited than a narrative review to estimate an overall effect and, critically, to test whether specific moderators systematically account for the observed heterogeneity.

Our focus on speech-LLM chatbots is warranted for both pedagogical and technological reasons. Pedagogically, oral communication poses challenges that text-based tools cannot fully address: learners need real-time practice in listening, speaking, and turn-taking under the temporal pressure of spoken interaction – opportunities frequently scarce in conventional classrooms (Shadiev and Liu 2023). By combining ASR with the conversational capacity of LLMs, speech-LLM chatbots offer a private, on-demand interlocutor with whom learners can rehearse spoken output repeatedly, free of the interpersonal anxiety often associated with speaking to peers or teachers (Tai and Chen 2025). Moreover, the extant research on AI in language learning remains disproportionately geared toward reading and writing skills (Law 2024; Liang et al. 2023), despite oral communication being highlighted as a crucial area for investigation (Jeon et al. 2024; Li et al. 2025). Technologically, the rapid maturation of speech interfaces in mainstream LLM products (e.g., voice-mode ChatGPT and Gemini) has shifted chatbot interaction from typing toward talking, making the spoken channel the fastest-growing yet least synthesized modality in this literature.

Our restriction to L2 learners follows the same rationale: the problems these tools address (e.g., limited speaking practice, communication apprehension, and restricted access to interlocutors) are specific to second- and foreign-language contexts, where most primary studies are situated; including L1 participants would conflate distinct learning processes and populations. Accordingly, we narrowed the scope of the meta-analysis to research from 2022 onward, which marked the emergence of LLMs in the TL literature, and specifically to the role of chatbots as

conversational partners, which remains their most extensively researched function. Regarding the latter, we focused exclusively on studies in which learners engaged in interactive oral communication tasks with chatbots in the TL and, consequently, excluded studies in which the chatbot's role was limited to providing one-way feedback on pronunciation. Through a highly focused synthesis of targeted findings, this meta-analysis offers a concentrated empirical perspective, enabling the development of nuanced pedagogical implications and concrete instructional frameworks for employing speech-LLM chatbots in TL teaching and learning. To this end, this study addresses the following two research questions:

RQ1: What is the overall effect of engaging in oral communication with speech-LLM chatbots on L2 learners' learning outcomes and affective variables?

RQ2: What variables (i.e., learner characteristics and intervention features) significantly moderate the effects of speech-LLM chatbots on L2 learning outcomes and affective variables?

2 Methods

2.1 Literature search

Studies were considered suitable for inclusion if they met the following criteria (see Table 1): (a) examined learners' voice-based interactions with LLM-based chatbots; (b) employed an experimental or quasi-experimental design, including a comparison group; (c) reported sufficient details to compute effect sizes (e.g., sample sizes, means and standard deviations); and (d) were published in English. To ensure the relevance of recent LLM and GenAI developments, the publication period was limited to studies published from 2022 onward, and only peer-reviewed journal articles were included. A systematic literature search was conducted across the major databases (Web of Science, Scopus, ERIC, and PsycINFO) and supplemented by manual searches. The manual search comprised two components. First, we hand-searched the tables of contents (2022 onward) of five journals: *Computer Assisted Language Learning, ReCALL, Language Learning & Technology, System,* and *RELC Journal*. These journals were selected given that they are recognized venues for computer-assisted language learning scholarship. The journals' online archives were searched using the same search terms employed in the database search. Second, we conducted backward and forward reference searches (i.e., screening the reference lists of, and articles citing, the included studies and relevant reviews).

To execute this search effectively, the query was formulated across four key dimensions – intervention, target outcomes, learning context, and methodology – and structured as follows: TITLE-ABS-KEY ((“chatbot*” OR “generative artificial intelligence”

Table 1: Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Type of technology	LLM/GenAI	Non-LLM/Non-GenAI
Interaction modality	Voice-based interaction or multimodal interaction (e.g., voice and text)	Text-only interaction
Study design	Experimental or quasi-experimental studies with a comparison group	Qualitative or quantitative studies without a comparison group
Outcome reporting	Sufficient statistical data to compute effect sizes	Lack of sufficient statistical data to compute effect sizes
Language	Written in English	Not written in English
Publication period	Published from 2022 onward	Published prior to 2022

OR "generative AI" OR "GenAI" OR "LLM*" OR "large language model*" OR "conversational agent*" OR "ChatGPT" OR "Gemini" OR "Google Assistant" OR "Alexa") AND ("speak*" OR "oral" OR "pronunciation") AND ("second language" OR "foreign language" OR "ESL" OR "EFL") AND ("quasi-experiment*" OR "experiment*" OR "random*" OR "control group" OR "experimental group" OR "posttest" OR "intervention")).

The rationale for this keyword combination is fourfold. First, to comprehensively capture the intervention (speech-enabled LLMs and chatbots), the query combined broad technological umbrella terms with specific commercial applications. Second, the target learning outcomes were restricted to oral proficiency by utilizing terms related to spoken language and articulation. Third, the educational context was strictly defined using terminology and acronyms specific to L2 to effectively exclude first language (L1) studies. Finally, methodological keywords were incorporated to isolate empirical studies with experimental or quasi-experimental designs, ensuring that the retrieved literature provides the quantitative data requisite for meta-analytic modeling.

Figure 1 illustrates the study's PRISMA-based screening process. In the initial identification stage, 194 records were retrieved – 121 from databases, 56 from manual searches of the target journals, and 17 from backward/forward reference searches. After removing 55 duplicates, 139 studies were retained for screening. Based on title and abstract screening, 4 studies were excluded as not relevant to the topic of interest, leaving 135 studies for the full-text review. Subsequently, based on the inclusion and exclusion criteria, these full texts were carefully evaluated for eligibility. Ultimately, 116 studies were excluded. The primary reasons for exclusion included the following: no focus on AI-learner interaction (n = 48), no LLM/GenAI involved (n = 27), absence of a quasi-experimental/experimental design (n = 23), no

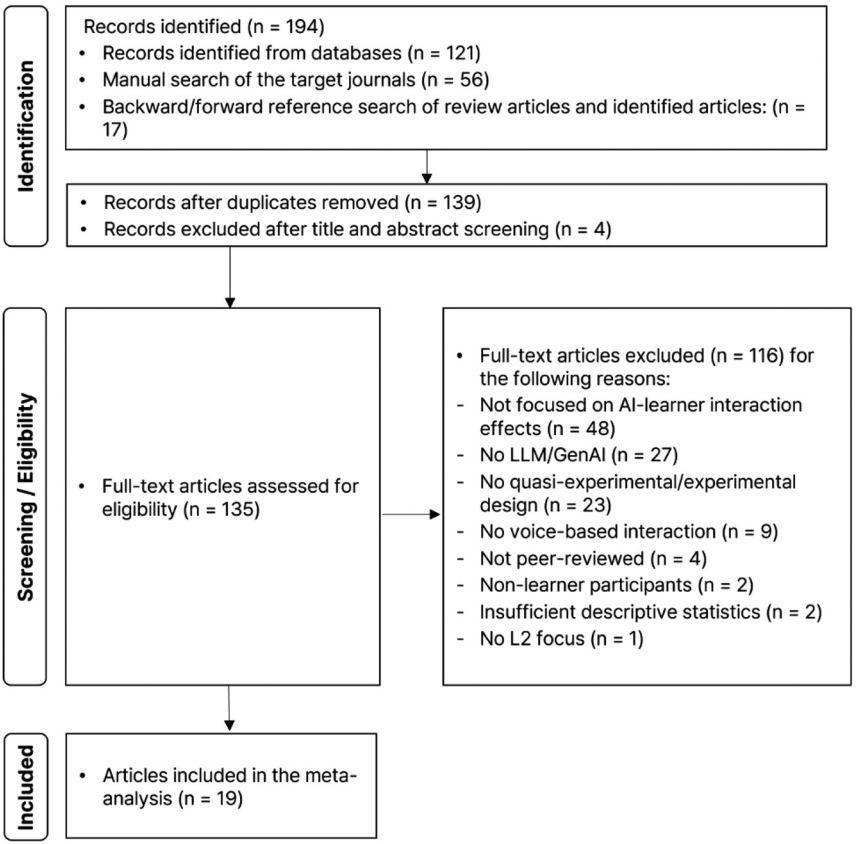


Figure 1: PRISMA flow diagram of literature search and study selection.

voice-based interactions (n = 9), not peer-reviewed articles (n = 4), non-learner participants (n = 2), insufficient descriptive statistics (n = 2), and lack of an L2 focus (n = 1). Furthermore, for the two studies with insufficient descriptive statistics, the authors were contacted to request the missing data, but no responses were received. Finally, 19 studies met all criteria and were included in the current meta-analysis.

2.2 Dataset construction

2.2.1 Understanding data structure

The studies included in this meta-analysis were examined to determine the structures of the datasets and ensure observation independence. The extracted data were

organized using a hierarchical framework. First, the independent samples within individual studies were identified; if a study featured multiple experimental conditions or distinct participant groups, each was treated as a separate, independent sample. Tai and Chen (2024) reported results for individual interactions (I-Bot) and pair interactions (P-Bot), which were coded as two independent samples. Similarly, Sonsaat-Hegelheimer and Kurt (2024) included two experimental groups that used different chatbot systems (Pronounce and Gemini), but only the group employing the LLM system was included in the present analysis.

Furthermore, when studies assessed the L2 learning outcomes and affective domains as dependent variables, they were treated as distinct, independent samples. Wu et al. (2026) utilized two instruments for the affective domain and a speaking test rated on four criteria for L2 learning outcomes, which were coded as two independent samples – Wu et al. (2026) [1] and Wu et al. (2026) [2]. Following this procedure, the dataset was structured hierarchically, with independent samples nested within studies. The final dataset comprised 32 independent samples (N = 2,643), derived from 19 studies.

2.2.2 Estimating effect sizes

Building upon the data structure described earlier, the effect sizes were calculated for each independent sample. Given the generally small sample sizes, Hedges' unbiased estimate of the standardized mean difference (Hedges' g) was applied to mitigate potential bias. The effect sizes represent the magnitudes of the educational impact observed in the experimental groups compared to the comparison groups (i.e., those that did not engage in communication with speech-LLM chatbots). Effect sizes were computed from post-test scores of the experimental and comparison groups (i.e., the standardized difference between group post-test means, using post-test standard deviations), rather than from gain scores or covariate-adjusted post-test means, because post-test descriptive statistics were the information most consistently reported across the primary studies. For affective outcomes, scores on negatively worded measures (anxiety) were reverse-coded so that positive effect sizes always indicate a more favorable affective state.

The dependent variables were L2 learning outcomes or affective domains; a positive effect size thus indicated improved L2 skills or a favorable affective shift. We define the affective domain as the set of learner emotions, attitudes, and dispositions toward L2 learning and communication measured in the primary studies, primarily comprising foreign language speaking anxiety, foreign language enjoyment, motivation, engagement, self-efficacy/self-confidence, and willingness to communicate. Although conceptually distinct, these constructs were pooled into a single affective domain. The primary reason is practical: too few effect sizes were available for

any single construct to support separate models, and pooling yields a more stable estimate of the overall affective impact, with the multilevel structure accommodating dependency among measures from the same sample. Control–Value Theory (Pekrun 2006, 2024) does not license treating these constructs as a single construct; it explicitly distinguishes control appraisals (self-efficacy/self-confidence), achievement emotions (anxiety, enjoyment), and their motivational and behavioral consequences (motivation, engagement, willingness to communicate; Dewaele and MacIntyre 2014; MacIntyre et al. 1998; Pekrun and Linnenbrink-Garcia 2012). What it does provide is grounds for regarding them as a theoretically related set of outcomes rather than an arbitrary collection of measures. The pooled effect size should accordingly be interpreted as the average effect across this set of related affective–motivational outcomes, not as an estimate of a unitary affective construct (see the second limitation in Section 5).

In this meta-analysis, multiple dependent variables were frequently measured for a single sample. In the two independent samples of Wu et al. (2026), the affective domain sample included measures for Foreign Language Speaking Anxiety (FLSA) and Foreign Language Enjoyment (FLE), generating two effect sizes. For the speaking proficiency sample, four distinct aspects (i.e., pronunciation, fluency and coherence, lexical resource, and grammatical range and accuracy) were assessed, yielding four effect sizes. In total, six effect sizes were derived from Wu et al. (2026). This resulted in a three-level hierarchical structure, with the sampling variance of each effect size at Level 1 and effect sizes (Level 2) nested within samples (Level 3). This multilevel framework accounted for the dependency among effect sizes derived from the same sample while maintaining independence across samples (Lee et al. 2019). Consequently, the final dataset comprised 77 effect sizes from 32 independent samples, as the basis for the multilevel meta-analysis.

2.2.3 Coding moderator variables

Each moderator and the corresponding categories are presented in Table 2. The moderator variables were categorized into two overarching groups, learner-related and intervention-related variables, in line with previous syntheses of chatbot-assisted language learning (Jeon et al. 2024; Li et al. 2025; Wang et al. 2025); both the variables and their categories (i.e., the coding scheme in Table 2) were selected on this basis prior to the analyses and were not derived from the framework in Section 1.1, which was applied post hoc to interpret them (see Section 1.1 for the mechanisms linking each moderator to that framework). Given the limited number of studies and the small samples, most moderator variables were coded in a binary manner to ensure statistical stability. All moderator variables were coded by two of the three authors using a shared coding manual with explicit category definitions. They first

independently coded 5 of the 19 included studies, yielding a Cohen's kappa of 0.84. Discrepancies in this calibration set were resolved through discussion. Then, they each coded half of the remaining studies; cases either coder flagged as ambiguous were resolved through discussion.

For learner-related variables, the education level was categorized into (a) primary and secondary (e.g., Kooti et al. 2025; Tai and Chen 2025) and (b) tertiary and others (e.g., Kang 2025; Shi and Shakibaei 2025). The latter category included studies with participants beyond the university level – adult learners ranging from 18 to 35 years old. L2 proficiency was categorized using three groups: beginner (e.g., Lee 2025; Zhang et al. 2026), intermediate (e.g., Guo et al. 2026; Xodabande et al. 2025) and mixed/others (e.g., Du and Reynolds 2025; Zheng et al. 2026).

For intervention-related variables, the type of chatbot was categorized as generic LLM (e.g., Kooti et al. 2025) or customized chatbot (e.g., Ding and Yusof 2025; Guan et al. 2025). Interaction modes were coded along two dimensions: (a) interaction format: individual (e.g., Shi and Shakibaei 2025; Zhang et al. 2024a) versus group-based interaction (e.g., Zhou et al. 2025) and (b) modality: oral-only (e.g., Kooti et al. 2025) versus oral-and-text interaction (e.g., Celik et al. 2025; Ding and Yusof 2025). Intervention length was dichotomized into shorter (less than a month; e.g., Sonsaat-Hegelheimer and Kurt 2024; Tai and Chen 2024) and longer durations (one to six months; e.g., Celik et al. 2025; Lee 2025). The chatbot use context was classified into three categories: in-class only (e.g., Lee 2025; Wu et al. 2026), out-of-class (e.g., Du and Reynolds 2025; Guan et al. 2025), and combined in- and out-of-class (e.g., Kooti et al. 2025; Zhou et al. 2025). Finally, the task type was coded as either freely used by learners (e.g., Zhang et al. 2024a) or pedagogically guided by instructors (e.g., Celik et al. 2025; Wu et al. 2026).

Table 2: Coding moderator variables.

Domain	Moderator	Category
Learner-related variables	Education level	Primary/secondary, tertiary/others
	L2 proficiency	Beginner, intermediate, mixed/others
Intervention-related variables	Type of chatbot	Generic LLM, customized chatbot
	Interaction format	Individual, group
	Modality	Oral only, oral-and-text
	Intervention length	Shorter, longer
	Context of chatbot use	In- and out-of-class, in-class only, out-of-class
	Type of task	Freely used, pedagogically guided
Methodology-related variable	Homogeneity confirmation	Established, unestablished

Additionally, to serve as a control variable in the meta-regression models, homogeneity confirmation of the experimental and comparison groups prior to the intervention based on pre-tests (or other measures related to L2 proficiency) was coded as either established (e.g., Du and Reynolds 2025; Kooti et al. 2025) or unestablished (e.g., Zhang 2026).

2.2.4 Publication bias

To assess the potential for publication bias, we created a funnel plot to visualize the distribution of effect sizes (see Figure 2). As subsequent data analyses (e.g., calculating overall effect sizes and conducting moderator analyses) were performed separately for two independent subgroups (19 samples [N = 1,334] and 36 effect sizes for L2 learning outcomes; 13 samples [N = 1,309] and 41 effect sizes for the affective domains), the publication bias analyses were also conducted twice.

The unit of analysis for publication bias was the sample rather than the individual effect size level. This approach was adopted because an outlying effect estimate at the sample level, rather than a single outlying effect size among several from the same sample, is more likely to reflect a bias inherent in the study's design or sampling (Viechtbauer and Cheung 2010). As illustrated in the funnel plots below, neither sample group exhibited distinct visual evidence of publication bias. Furthermore, we employed Egger's regression test to check whether the visual signs of the funnel plot were a significant indication of publication bias among effect sizes (Sterne et al. 2011). Egger's test was not statistically significant for either subgroup (p = 0.151 and 0.068, respectively), indicating that no statistically significant evidence of publication bias was detected. Given the small number of samples, however, these tests are likely underpowered, and the absence of significant asymmetry should not be read as positive evidence that publication bias is absent.

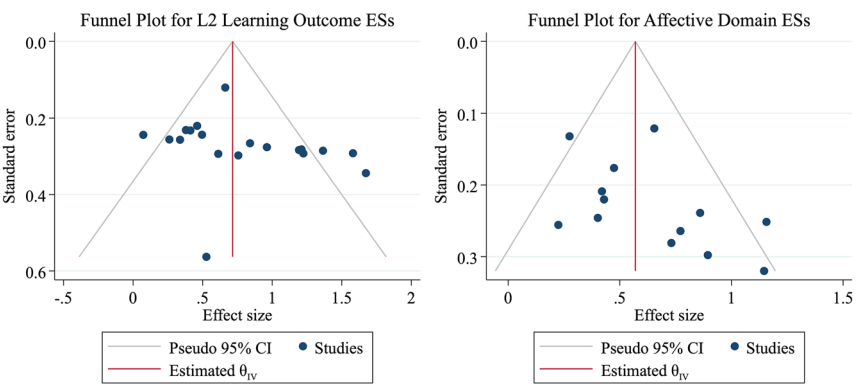


Figure 2: Two funnel plots for samples for L2 learning outcomes (left) and affective domains (right).

2.3 Dataset analysis

We conducted a multilevel meta-regression analysis to evaluate the overall effect sizes for engaging in oral communication with speech-LLM chatbots on L2 learning outcomes and affective domains, and to explore the moderating roles of learner characteristics and intervention features. All statistical procedures were performed using Stata, version 16.

To calculate the overall effect sizes, we first treated the computed effect sizes as the dependent variable and used their associated standard errors to define Level 1 variance (Lee et al. 2019). In this unconditional model, the intercept corresponds to the overall mean effect size.

Subsequently, to conduct moderator analyses, a comprehensive multiple meta-regression approach was employed. Initially, a full model incorporating all potential moderators was tested. To increase the validity of our statistical inferences, an indicator of baseline homogeneity between the experimental and control groups was included as a control variable in the model. Finally, to construct a parsimonious model and prevent overfitting, a reduced model was fitted. Given the relatively small number of included studies and the exploratory nature of the moderator analysis, variables that emerged as statistically significant ($p < 0.05$) or marginally significant ($p < 0.10$) in the full model were retained for this final analysis to minimize the risk of Type II errors.

3 Results

3.1 RQ1: overall effectiveness

3.1.1 L2 learning outcomes

Figure 3 displays the forest plot for effect sizes for the 19 samples. Although six samples included zero in the 95 % confidence intervals (i.e., there was no significant difference between the speech-LLM chatbot and comparison conditions), the rest showed positive effect sizes, ranging from 0.46 to 1.67, which point to the superiority of the former. The multilevel meta-analysis of effect sizes further revealed that the overall effectiveness of engaging in oral communication with speech-LLM chatbots, compared to the traditional learning condition, was 0.68 ($SE = 0.11$, $z = 5.94$, $p < 0.001$, 95 % CI [0.45, 0.90]).

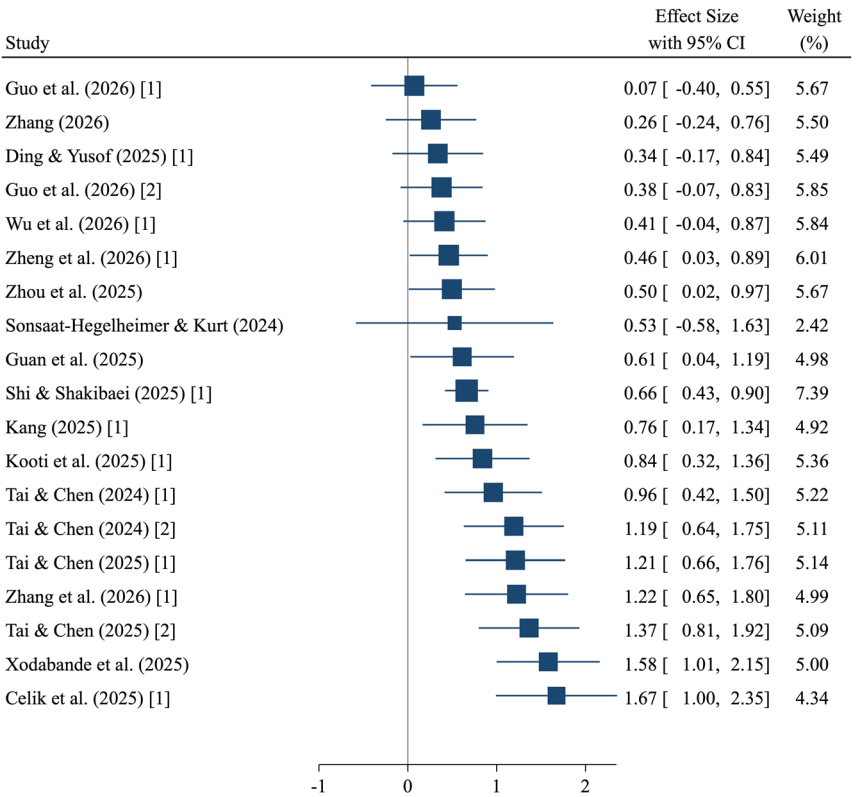


Figure 3: Forest plot for effect sizes on L2 learning outcomes at the sample level (k = 19).

3.1.2 Affective domains

Figure 4 displays the forest plot for effect sizes for 13 samples. Although three samples included zero in the 95 % confidence intervals, the rest showed positive effect sizes, ranging from 0.27 to 1.16, showing the superiority of the experimental condition. The multilevel meta-analysis of effect sizes further revealed that the overall effectiveness of engaging in oral communication with speech-LLM chatbots in terms of affective dimensions was 0.64 (SE = 0.08, z = 8.28, p < 0.001, 95 % CI [0.49, 0.79]).

3.2 RQ2-1: moderating effects on L2 learning outcomes

To address the second research question, a comprehensive multiple meta-regression approach was applied to examine the moderating effects of various factors on L2

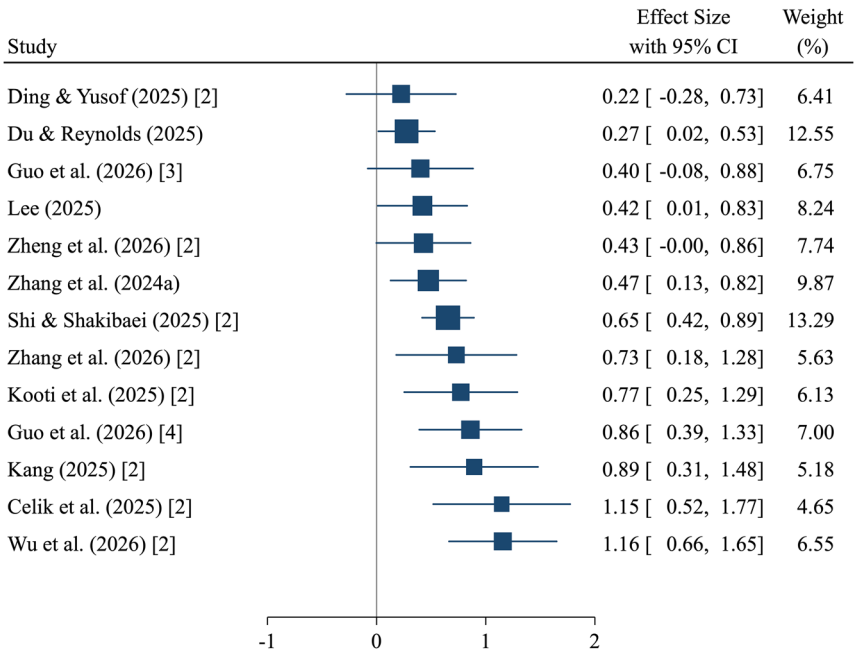


Figure 4: Forest plot for effect sizes on affective domains at the sample level (k = 13).

learning outcomes. Before presenting these results, we note that the number of studies, samples, and effect sizes available for the moderator analyses was small, relative to the number of moderators considered (see Appendix for the distribution of studies, samples, and effect sizes across the levels of each moderator). Because several moderator categories are represented by only a few samples, the moderator estimates below should be regarded as exploratory and interpreted with caution. Initially, the full model (Table 3) identified several variables with significant ($p < 0.05$) or marginally significant moderating effects ($p < 0.10$): L2 proficiency (beginner and intermediate levels), context of chatbot use, and type of task.

Subsequently, a parsimonious model was fitted retaining these variables alongside the baseline homogeneity control (Table 4). In the final model, only L2 proficiency emerged as a statistically significant moderator. Specifically, learners with beginner-level L2 proficiency demonstrated a significantly larger effect size (Coeff. = 0.91, SE = 0.27, $z = 3.34$, $p = 0.001$) compared to the reference group (Mixed/others). The remaining variables did not show statistically significant moderating effects in the reduced model ($p > 0.05$).

Table 3: L2 learning outcomes: Full multiple meta-regression model including all potential moderators.

Moderator	Category	Coeff.	SE	*z*	*p*
Educational level	Tertiary/others	0.57	0.37	1.55	0.122
	Primary/secondary		(Reference)		
L2 proficiency	Beginner	1.51**	0.51	2.94	0.003
	Intermediate	0.51†	0.28	1.80	0.071
	Mixed/others		(Reference)		
Type of chatbot	Generic LLM	0.14	0.24	0.60	0.547
	Customized chatbot		(Reference)		
Interaction format	Individual	0.04	0.20	0.22	0.827
	Group		(Reference)		
Modality	Oral only	0.22	0.26	0.83	0.405
	Oral-and-text		(Reference)		
Intervention length	Longer	−0.04	0.24	−0.18	0.860
	Shorter		(Reference)		
Context of chatbot use	In- and out-of-class	0.45†	0.24	1.85	0.065
	In class/out-of-class		(Reference)		
Type of task	Freely used	−0.40†	0.22	−1.84	0.066
	Pedagogically guided		(Reference)		
Homogeneity confirmation	Established	0.12	0.23	0.52	0.600
	Unestablished		(Reference)		

The reference level of each moderator variable is marked "(reference)". Coefficients are estimated relative to the marked reference level. $^{**}p < 0.01$, $^{\dagger}p < 0.10$.

3.3 RQ2-2: moderating effects on affective domains

A parallel multiple meta-regression analysis was conducted for the affective domains. The full model (Table 5) indicated that three variables were statistically

Table 4: L2 learning outcomes: Final parsimonious multiple meta-regression model with significant moderators.

Moderator	Category	Coeff.	SE	*z*	*p*
L2 proficiency	Beginner	0.91**	0.27	3.34	0.001
	Intermediate	0.34	0.25	1.35	0.175
	Mixed/others		(Reference)		
Context of chatbot use	In- and out-of-class	0.31	0.21	1.48	0.140
	In class/out-of-class		(Reference)		
Type of task	Freely used	−0.35	0.22	−1.63	0.103
	Pedagogically guided		(Reference)		
Homogeneity confirmation	Established	0.07	0.17	0.37	0.709
	Unestablished		(Reference)		

The reference level of each moderator variable is marked "(reference)". $^{**}p < 0.01$.

Table 5: Affective domains: Full multiple meta-regression model including all potential moderators.

Moderator	Category	Coeff.	SE	*z*	*p*
Educational level	Tertiary/others	0.61*	0.24	2.53	0.011
	Primary/secondary		(Reference)		
L2 proficiency	Beginner	0.33	0.24	1.39	0.164
	Intermediate	0.25	0.16	1.57	0.115
	Mixed/others		(Reference)		
Type of chatbot	Generic LLM	−0.15	0.25	−0.60	0.550
	Customized chatbot		(Reference)		
Interaction format	Individual	−0.50*	0.25	−2.01	0.044
	Group		(Reference)		
Modality	Oral only	0.24	0.24	0.99	0.321
	Oral-and-text		(Reference)		
Intervention length	Longer	−0.21	0.34	−0.63	0.531
	Shorter		(Reference)		
Context of chatbot use	In- and out-of-class	0.70*	0.28	2.51	0.012
	In class/out-of-class		(Reference)		
Type of task	Freely used	0.02	0.16	0.14	0.888
	Pedagogically guided		(Reference)		
Homogeneity confirmation	Established	−0.18	0.31	−0.58	0.561
	Unestablished		(Reference)		

The reference level of each moderator variable is marked "(reference)". Coefficients are estimated relative to the marked reference level. *$p < 0.05$.

significant moderators ($p < 0.05$): educational level, interaction format, and context of chatbot use.

These variables were retained in the final parsimonious model (Table 6), where all three maintained their statistical significance. Specifically, the effectiveness of

Table 6: Affective domains: Final parsimonious multiple meta-regression model with significant moderators.

Moderator	Category	Coeff.	SE	*z*	*p*
Educational level	Tertiary/others	0.54*	0.22	2.50	0.010
	Primary/secondary		(Reference)		
Interaction format	Individual	−0.54*	0.21	−2.57	0.010
	Group		(Reference)		
Context of chatbot use	In- and out-of-class	0.74***	0.19	3.92	<0.001
	In class/out-of-class		(Reference)		
Homogeneity confirmation	Established	−0.23	0.12	−1.92	0.054
	Unestablished		(Reference)		

The reference level of each moderator variable is marked "(reference)". *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

speech-LLM chatbots on affective domains was significantly larger for learners in tertiary education (Coeff. = 0.54, SE = 0.22, z = 2.50, p = 0.01) compared to those in primary and secondary education. Furthermore, combined in- and out-of-class chatbot use yielded a significantly larger effect size (Coeff. = 0.74, SE = 0.19, z = 3.92, p < 0.001) than use restricted to solely in-class or out-of-class settings. Conversely, engaging with the chatbot in an individual interaction format resulted in a significantly smaller effect size (Coeff. = −0.54, SE = 0.21, z = −2.57, p = 0.01) compared to group-based interactions.

4 Discussion

This meta-analysis identified studies on the effects of engaging in oral communication with speech-LLM chatbots and synthesized their findings using a multilevel meta-analytic approach. This section first discusses the findings on the overall effectiveness of these tools, followed by the results of moderator analyses concerning learner characteristics and intervention features.

4.1 Overall effectiveness of engaging in oral communication with speech-LLM chatbots

The results for the overall effectiveness revealed that speech-LLM chatbots had a medium-sized effect (i.e., between-groups) on L2 learners (g = 0.68, SE = 0.11, z = 5.94, p < 0.001 for L2 learning outcomes and g = 0.64, SE = 0.08, z = 8.28, p < 0.001 for affective domains). These findings match those of Lyu et al.'s (2025) recent meta-analysis on chatbots for language learning, which also considered learning outcomes (g = 0.590) and affective dimension (g = 0.645), as well as those of extant studies that have focused on language skill developments (Li et al. 2025, g = 0.576; Wu and Li 2026, g = 0.648; Zhang et al. 2023, g = 0.527).

The positive findings for speech-LLM chatbots revealed in our meta-analysis results may be the result of the combined effects of recent advancements in automatic speech recognition (Oh and Song 2021; Shadiev and Liu 2023) and LLM/GenAI's superior capacity for contextual dialogue and authentic conversation (Lee et al. 2024; Li et al. 2025; Moorhouse 2024), both lending greater authenticity (Zhang et al. 2024b) to the user experience when conversing with chatbots. This enhanced authenticity successfully activates the psychological affordances of GenAI by lowering learners' anxiety levels during L2 practice (Kooti et al. 2025; Wu et al. 2026) and enhancing their motivation and self-confidence (Du and Reynolds 2025; Shi and Shakibaei 2025). These features position cutting-edge speech-LLM chatbots as ideal

conversational partners and engaging resources for L2 practice (Jeon et al. 2024), likely accounting for the gains observed in the learning and affective dimensions.

Meanwhile, a review of the included studies revealed variations in learner characteristics and intervention features, identified as potential variables that may moderate the effects of the target technology. The following two subsections discuss the findings related to the statistically significant moderator variables.

4.2 Moderating effects of learner-related variables

Among the examined learner characteristics, two moderators were found to significantly influence the overall effectiveness of engaging with speech-LLM chatbots. Regarding L2 learning outcomes, a statistically significant moderator was learners' L2 proficiency, with beginner-level learners showing a larger effect size than those at other proficiency levels. This finding contrasts with Wang et al. (2025), in which intermediate learners were found to benefit to a greater extent than their beginner counterparts. Regarding the potential reasons behind these outcomes, Wang et al. (2025: 647) suggest that learners "who have acquired basic language knowledge tend to have an enjoyable learning experience" and "experience less communication breakdown in the automatic speech recognition system in voice chats with chatbots." Meanwhile, they further suggested that if beginner-level learners could overcome such communication breakdowns, they might also benefit from chatbots. Since LLM chatbots are generally more effective at preventing such communication breakdowns (Lee et al. 2024; Li et al. 2025; Moorhouse 2024), the condition Wang et al. hypothesized can be seen as realized in the present study, which may explain why beginner-level learners exhibited large gains when engaging with speech-LLM chatbots. From an interactionist standpoint, this finding may be attributed to the adaptive, repair-capable output of LLMs, which sustains comprehensible interaction precisely for the learners who would otherwise face the most frequent breakdowns, maximizing opportunities for negotiated meaning at the beginner level (Long 1996).

Regarding affective outcomes, educational level emerged as a significant moderator, with learners in tertiary education benefiting to a greater extent than those at the primary and secondary levels. Because our affective measure pooled conceptually distinct constructs (e.g., motivation, self-efficacy, anxiety, and willingness to communicate), this moderation effect likely reflects several complementary mechanisms. With respect to the emotional component (anxiety), Control–Value Theory suggests that tertiary learners' greater autonomy and self-regulatory capacity plausibly strengthen both control appraisals (confidence in managing the interaction and

the technology) and value appraisals (recognition of the tool's relevance to their goals) – the two proximal antecedents of achievement emotions (Pekrun 2006, 2024). With respect to the motivational components, these same capacities may allow tertiary learners to leverage the technology for practical applications and personalized support (Wang et al. 2025), and such autonomy-driven engagement may in turn foster greater motivation and willingness to communicate.

4.3 Moderating effects of intervention-related variables

Two variables related to the implementation of speech-LLM chatbots moderated their effectiveness in the affective domain. First, studies implementing chatbots in group settings demonstrated a larger effect size in the affective domain than those in which participants engaged with the chatbot individually. A plausible explanation is that learners, particularly younger or lower-proficiency ones, experience less anxiety when tackling chatbot tasks collaboratively: as they share the conversational load and are not pressured to produce an utterance at every turn (Tai and Chen 2024), the collaborative environment likely fosters a more positive emotional experience. Notably, this finding refines, rather than confirms, the assumption articulated in Section 1.2 that the affective benefit of speech-LLM chatbots derives primarily from their nature as a private, individual interlocutor. The results suggest instead that when peers share responsibility for sustaining the conversation, the communicative demand on any single learner is lowered, which may bolster each learner's control appraisal, while the social setting itself may add enjoyment – an achievement emotion that Control–Value Theory ties closely to the perceived intrinsic value of the activity (Pekrun 2006, 2024) and that has been shown to carry a strong social dimension in L2 classrooms (Dewaele and MacIntyre 2014). However, this aspect of the intervention has received limited attention in chatbot studies. As such, more research is needed to explore the dynamics of peer collaboration during speech-LLM chatbot tasks and how they influence learners' affective outcomes.

The other statistically significant moderator in the affective domain concerned the context of chatbot use. The finding that combined use of chatbots across in-class and out-of-class settings yielded a larger effect size than either setting alone is generally consistent with previous meta-analyses (e.g., Lyu et al. 2025; Wang et al. 2025), although these differences did not reach significance in those syntheses. Encouraging learners to utilize chatbots beyond the classroom capitalizes on one of their key pedagogical affordances – availability without temporal or spatial constraints (Huang et al. 2022). Our interpretive framework offers a complementary reading: combined deployment increases the sheer quantity of meaningful oral

interaction available to learners (the interactionist mechanism), while out-of-class access grants learners discretion over when, where, and for how long they practice – autonomy-supportive conditions that Control–Value Theory identifies as environmental antecedents of heightened control appraisals and, through them, more favorable achievement emotions (Pekrun 2006, 2024). Practitioners should therefore strategically integrate out-of-class chatbot assignments to reinforce in-class instruction, creating a continuous, LLM-supported L2 learning experience.

5 Conclusions

Although technological advances have steadily enhanced chatbots' conversational capacity, the integration of LLMs has expanded it substantially. These chatbots have been shown to offer an authentic experience of communicating in the TL and facilitate a wider range of L2 conversation tasks. Accordingly, this meta-analysis aimed to synthesize the findings of studies on the effects of engaging in L2 oral communication with speech-LLM chatbots. Overall, our findings demonstrated that L2 learners benefit from engaging in oral communication with these tools, in terms of L2 learning outcomes and, to a similar degree, across affective dimensions. Given the importance of affective factors and their sustained impact on L2 learning, these findings provide empirical support for integrating speech-LLM chatbots into L2 curricula and programs.

These findings advance the literature in ways that carry concrete pedagogical implications. Whereas prior syntheses have largely pooled voice- and text-mode or pre-LLM chatbots, the present study provides quantitative evidence specific to LLM-based spoken interaction, filling a clearly identified gap. For practitioners, the results suggest that speech-LLM chatbots are a viable means of expanding scarce TL speaking practice, and that their affective benefits – larger for tertiary learners, group-based use, and blended in- and out-of-class deployment – can be leveraged by embedding chatbot tasks across classroom and self-access settings and, where feasible, in collaborative configurations. Viewed through the interpretive framework adopted here, these patterns favor designs that maximize meaningful oral interaction while sustaining learners' sense of control. Because several moderator effects rest on limited samples, however, they are best treated as promising directions for instruction and confirmatory research rather than settled prescriptions.

This study is not without limitations. First, the number of included studies and their corresponding sample sizes were relatively small. Our analyses detected no statistically significant evidence of publication bias in the current dataset, although such tests are likely underpowered given the small number of samples. Given the

rapid proliferation of LLM and GenAI research on the use of chatbots for L2 teaching and learning, future meta-analyses can incorporate a broader corpus of studies, increasing the statistical power of the findings. Second, because the framework in Section 1.1 was applied post hoc, the mechanisms it proposes are interpretive hypotheses awaiting direct testing; the consistency between the present findings and the framework should therefore not be read as confirmatory evidence for it. A further constraint on testing the framework is that the affective and motivational variables were merged into a single domain. Although necessitated by the small number of effect sizes per construct, this merging precludes construct-specific estimates; because these constructs occupy different positions in the appraisal–emotion–motivation chain, they may respond differently to chatbot interaction, and the pooled estimate should not be read as evidence of a single affective construct. As the literature accumulates, future syntheses should model individual constructs separately to test whether the benefits observed here hold across the affective domain.

In conclusion, this meta-analysis provides a critical empirical foundation for the pedagogical shift toward LLM/GenAI-mediated oral practice using chatbots. As researchers continue to refine methodological reporting and explore diverse instructional contexts, speech-LLM chatbots are expected to become a promising component of innovative L2 learning environments.

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Appendix: Number of studies (*k*), independent samples, and effect sizes per moderator category

Moderator	Category	k (studies)	Samples	Effect sizes
Education level	Primary/secondary	4	7	10
	Tertiary/others	15	25	67
L2 proficiency	Beginner	4	7	18
	Intermediate	10	19	44
	Mixed/others	5	6	15
Type of chatbot	Generic LLM	10	18	52
	Customized chatbot	9	14	25
Interaction format	Individual	16	27	64
	Group	4	5	13
Modality	Oral only	5	8	16
	Oral-and-text	15	24	61
Intervention length	Shorter (<1 month)	5	8	13
	Longer (1–6 months)	14	24	64
Context of use	In-class only	8	13	30
	Out-of-class	6	11	33
	In- and out-of-class	5	8	14
Type of task	Freely used	3	5	21
	Pedagogically guided	17	27	56

k = number of studies; Samples = number of independent samples; Effect sizes = number of effect sizes. Counts were computed from the coded dataset; because a single study may contribute samples to more than one category, *k* across categories within a moderator may exceed the total number of studies. Note. The 19 studies included in the meta-analysis were Celik et al. (2025), Ding and Yusof (2025), Du and Reynolds (2025), Guan et al. (2025), Guo et al. (2026), Kang (2025), Kooti et al. (2025), Lee (2025), Shi and Shakibaei (2025), Sonsaat-Hegelheimer and Kurt (2024), Tai and Chen (2024, 2025), Wu et al. (2026), Xodabande et al. (2025), Zhang et al. (2024a), Zhang (2026), Zhang et al. (2026), Zheng et al. (2026), and Zhou et al. (2025).

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